Proximity to Parks and Asthma Prevalence in Chicago: A GIS Analysis of Environmental Access and Respiratory Health

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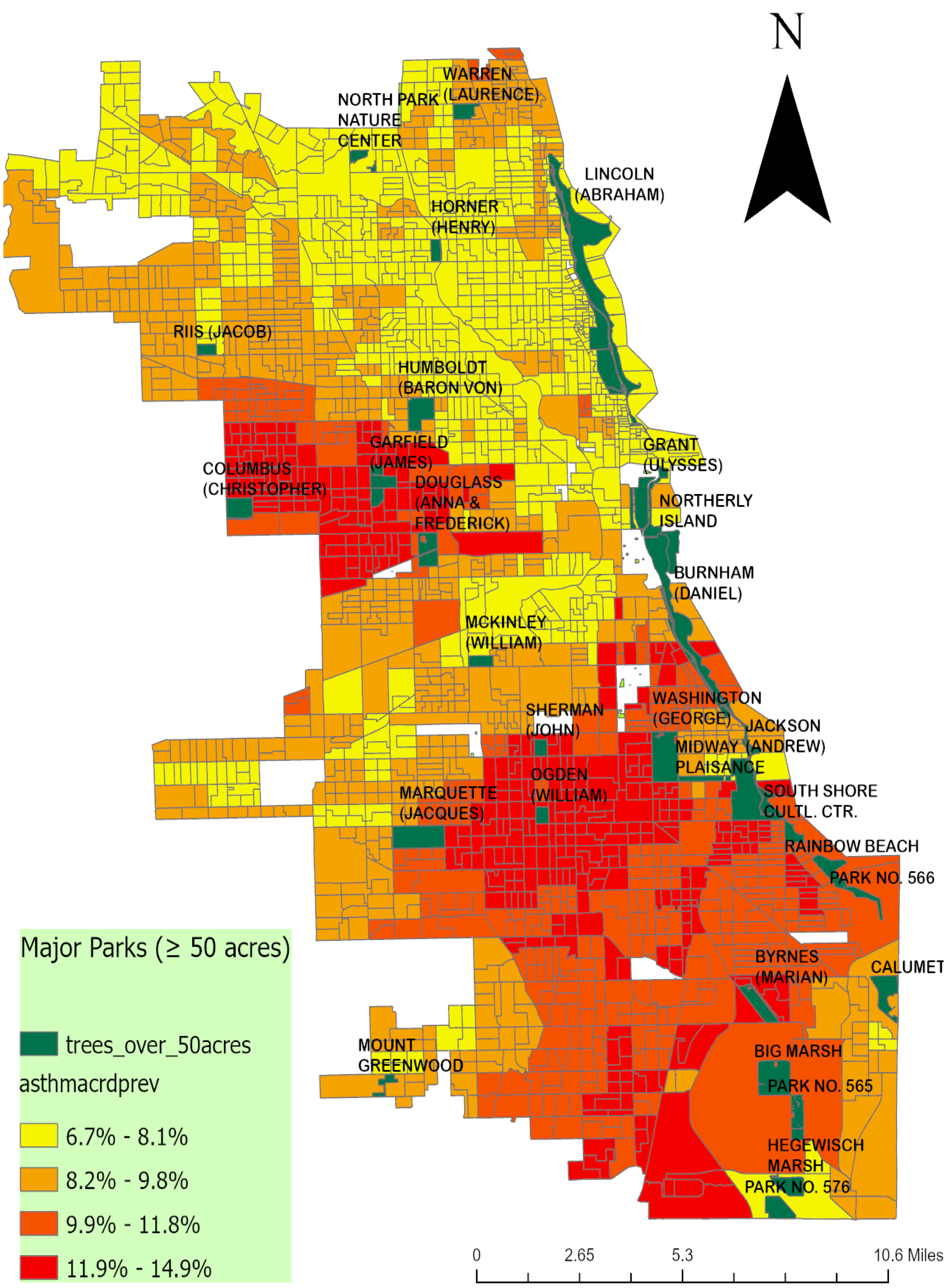
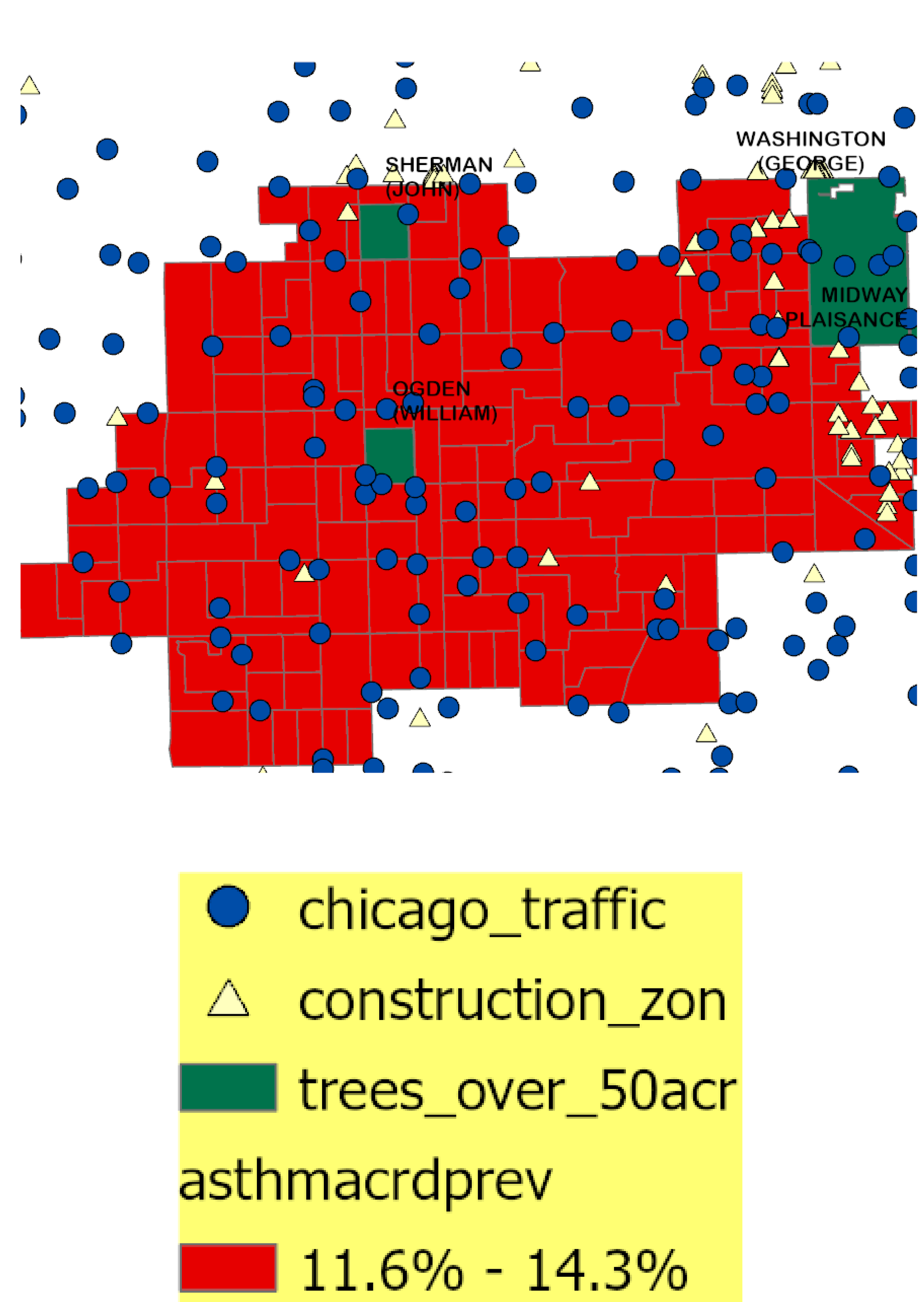


Intercept	6.231*** (0.057)
Below Poverty Line (%)	1.712*** (0.145)
Minority Population (%)	2.488*** (0.103)
Less than High School (%)	-0.953*** (0.223)
Linguistically Isolated (%)	-6.894*** (0.234)
R2	0.784
* p < 0.05, ** p < 0.01, *** p < 0.001	

Table: Regression Results: Social Indicators and Asthma Prevalence



Introduction

Asthma is a major respiratory health concern in Chicago. Access to green spaces such as parks may be associated with better respiratory health outcomes. Therefore, this project seeks to answer the following research question: **Is proximity to parks associated with higher or lower asthma prevalence across Chicago census block groups?**

Materials and Methods

Data were obtained from the Chicago Data Portal and Census.gov. Spatial analysis was conducted using table joins and spatial overlays at the census block group level. A choropleth map was created to display asthma prevalence across Chicago, overlaid with environmental data. In R, a multivariate regression was performed with asthma prevalence as the outcome variable.

Limitations

The results are based on census block group averages and cannot be interpreted at the individual level. Moreover, it reflects only spatial proximity to park boundaries rather than actual environmental benefits. Since the analysis is cross-sectional, the results identify associations rather than causal relationships between park proximity and asthma prevalence. Another key limitation would be Omitted Variable Bias.

Main Findings

The choropleth map shows asthma prevalence across Chicago census block groups, with darker areas indicating higher rates. High asthma prevalence clusters around Ogden, Washington, and Sherman Park, which also overlap with high traffic and construction activity. North and lakefront areas (near parks like Lincoln Park) tend to have lighter colors. Regression results further highlight the role of social vulnerability in spatial disparities. A one percentage increase in poverty rate is associated with a 1.71% increase in asthma prevalence, holding all variables constant.

Conclusion and Future Work

Areas with greater park boundary coverage generally tend to show lower asthma prevalence. But some census block groups with large parks still experience relatively high asthma rates. This suggests other factors are at play. Future work should incorporate data on park usage, accessibility, and air quality metrics (PM2.5) to understand the relationship with asthma prevalence.

Data Source

1. City of Chicago | Data Portal | City of Chicago | Data Portal. (n.d.). Chicago. https://data.cityofchicago.org/
2. US Census Bureau. (2025, December 23). TIGER/Line shapefiles. Census.gov. https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html